

Flatcar Deployments

Source section: `deploy/` (renamed from `installing/` in the refactored branch). Covers cloud providers, virtualization platforms, and bare metal.

Cloud Providers (`deploy/cloud/`)

Provider	Support
Amazon EC2	Official
Microsoft Azure	Official
Google Compute Engine	Official
VMware	Official
Hetzner	Official
DigitalOcean	Official
Akamai/Linode	Official
STACKIT	Official
Brightbox	Official
Exoscale	Community
Scaleway	Community (moved to <code>virt-options/</code> in this branch)
OracleCloud	Community (moved to <code>virt-options/</code>)
OVHcloud	Community (moved to <code>virt-options/</code>)
Vultr	Community (new in this branch)

Ignition/Butane config is passed via the provider's **user data** / **custom data** field at instance creation.

Virtualization Options (`deploy/virt-options/`)

In the refactored branch, `virt-options/` consolidates VMs and some community platforms that were split across `installing/vms/` and `installing/community-platforms/` in main:

Platform	Support
QEMU/KVM	Official
libvirt	Official

Platform	Support
KubeVirt	Community
Hyper-V	Community
VirtualBox	Community
Vagrant	Community
Proxmox VE	Community
OpenStack	Community (moved from cloud/ in this branch)
Eucalyptus	Community

KubeVirt (`deploy/virt-options/kubevirt.md`)

KubeVirt images are `qcow2` format for both `amd64` and `arm64` , available from Alpha 3975.0.0.

Download:

```
wget https://alpha.release.flatcar-linux.net/amd64-usr/3975.0.0/flatcar_production_kubevirt_image.qcow2
```

Upload via virtctl:

```
virtctl image-upload pvc flatcar-production-kubevirt-image --size=10Gi \
  --image-path=flatcar_production_kubevirt_image.qcow2 \
  --uploadproxy-url https://cdi-uploadproxy:31001 --insecure
```

Or via PVC with CDI import annotation:

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: flatcar-amd64-3975
  annotations:
    cdi.kubevirt.io/storage.import.endpoint: "https://alpha.release.flatcar-linux.net/amd64-usr/3975.0.0/flatcar_production_kubevirt_image.qcow2"
spec:
  accessModes: [ReadWriteOnce]
  resources:
    requests:
      storage: 10Gi
  storageClassName: ceph-block
```

VM with cloud-init userdata:

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: vm-flatcar-cfgdrive
spec:
  running: true
  template:
    spec:
      domain:
        cpu: {cores: 1}
        devices:
          disks:
            - {disk: {bus: virtio}, name: disk0}
            - {disk: {bus: sata}, name: cloudinitdisk}
          resources:
            requests: {memory: 1024M}
      volumes:
        - name: disk0
          persistentVolumeClaim:
            claimName: flatcar-amd64-3975
        - cloudInitConfigDrive:
            userData: |
              #!/bin/bash
              echo "core:foo" | chpasswd
            name: cloudinitdisk
```

VM with Ignition userdata (generate JSON from Butane first):

```
butane < userdata.yaml > userdata.json
```

```
# userdata.yaml (Butane)
variant: flatcar
version: 1.0.0
kernel_arguments:
  should_exist:
    - flatcar.autologin
passwd:
  users:
    - name: core
      password_hash: $6$...hash...
```

Then embed the JSON output as `cloudInitConfigDrive.userData` in the `VirtualMachine` spec using a `cdrom` disk (not `disk`).

Both cloud-init and Ignition are supported via `cloudInitConfigDrive`. KubeVirt is community-supported — no official release tests run for it.

Bare Metal (`deploy/bare-metal/`)

Method	Notes
ISO	Boot and install interactively
PXE	Network boot; must add <code>flatcar.first_boot=1</code> to kernel args
iPXE	Network boot via HTTP
<code>flatcar-install</code>	Run on existing Linux system to install to disk
Raspberry Pi	ARM64 support

PXE requirement: Since GRUB is not used, set `flatcar.first_boot=1` explicitly in kernel cmdline — do not use `detected`.

For bare metal metadata provisioning, use [Matchbox](#).

Notes for Distributors (`deploy/virt-options/notes-for-distributors.md`)

Covers Flatcar-specific considerations for platform vendors building images or integrations, including OEM partition usage and update server requirements.